Does advancing technology make society better? In some ways, perhaps. However, many films and TV shows—as far back as *Frankenstein* (*left*) in 1931 and as recently as the 2013 film *Iron Man 3* (*right*)—have expressed the concern that new technology not only solves old problems but also creates new ones. All the sociological theorists discussed in this chapter shared this ambivalent view of the modern world.

Postindustrial Societies

Many industrial societies, including the United States, have now entered a new phase of technological development, and we can extend Lenski's analysis to take account of recent trends. A generation ago, the sociologist Daniel Bell (1973) coined the term **postindustrialism** to refer to *the production of information using computer technology*. Production in industrial societies centers on factories and machinery generating material goods; postindustrial production relies on computers and other electronic devices that create, process, store, and apply information. Just as people in industrial societies learn mechanical skills, people in postindustrial societies such as ours develop information-based skills and carry out their work using computers and other forms of high-technology communication.

As Chapter 16 ("The Economy and Work") explains, a postindustrial society uses less and less of its labor force for industrial production, and the work can be performed almost anywhere. At the same time, more jobs become available for clerical workers, teachers, writers, sales managers, and marketing representatives, all of whom have in common jobs that involve processing information.

The Information Revolution, which is at the heart of postindustrial society, is most evident in rich nations, yet new information technology affects people in all countries around the world. As discussed in Chapter 3 ("Culture"), a worldwide flow of products, people, and information now links societies and has advanced a global culture. In this sense, the postindustrial society is at the heart of globalization.

The Limits of Technology

More complex technology has made life better by raising productivity, reducing infectious disease, and sometimes just relieving boredom. But technology provides no quick fix for social problems. Poverty, for example, remains a reality for some 45.3 million women and men in the United States (see Chapter 11, "Social Class in the United States") and just over 1 billion people worldwide (U.S. Census Bureau, 2014; World Bank, 2014; see Chapter 12, "Global Stratification").

Technology also creates new problems that our ancestors (and people like Sididi Ag Inaka today) could hardly imagine. Industrial and postindustrial societies give us more personal freedom, but they often lack the sense of community that was part of preindustrial life. Most seriously, an increasing number of the world's nations have used nuclear technology to build weapons that could send the entire world back to the Stone Age—if humanity survives at all.

Advancing technology has also threatened the physical environment. Each stage in sociocultural evolution has introduced more powerful sources of energy and increased our appetite for Earth's resources. Ask yourself whether we can continue to pursue material prosperity without permanently damaging our planet by consuming its limited resources or poisoning it with pollution (see Chapter 22, "Population, Urbanization, and Environment").

Technological advances have improved life and brought the world's people closer. But establishing peace, ensuring justice, and protecting the environment are problems that technology alone cannot solve.

Karl Marx: Society and Conflict

4.2 Analyze the importance of class conflict to the historical development of human societies.

The first of our classic visions of society comes from Karl Marx (1818–1883), an early giant in the field of sociology whose influence continues today. Keenly aware of how the

Industrial Revolution had changed Europe, Marx spent most of his adult life in London, the capital of what was then the vast British Empire. He was awed by the size and productive power of the new factories going up all over Britain. Along with other industrial nations, Britain was producing more goods than ever before, drawing raw materials from around the world and churning out finished products at a dizzying rate.

What astounded Marx even more was that the riches produced by this new technology ended up in the hands of only a few people. As he walked around the city of London, he could see for himself that a handful of aristocrats and industrialists enjoyed lives of luxury and privilege, living in fabulous mansions staffed by many servants. At the same time, most people lived in slums and labored long hours for low wages. Some even slept in the streets, where they were likely to die young from diseases brought on by cold and poor nutrition.

Marx saw his society in terms of a basic contradiction: In a country so rich, how could so many people be so poor? Just as important, he asked, how can this situation be changed? Many people think Marx set out to tear societies apart. But he was motivated by compassion and wanted to help a badly divided society create a new and more just social order.

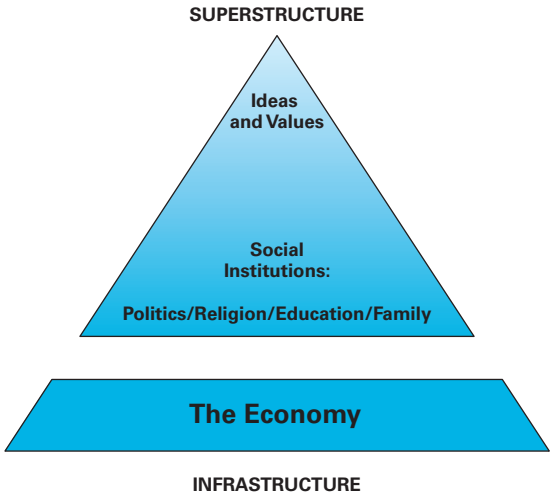


Figure 4–1 Karl Marx's Model of Society

This diagram illustrates Marx's materialist view that the system of economic production shapes the entire society. Economic production involves both technology (industry, in the case of capitalism) and social relationships (for capitalism, the relationship between capitalists, who own the factories and businesses, and workers, who provide labor). On this infrastructure, or foundation, rests society's superstructure, including its major social institutions as well as core cultural values and ideas. Marx maintained that every part of a society operates to support the economic system.

social conflict the struggle between segments of society over valued resources

capitalists people who own and operate factories and other businesses in pursuit of profits

proletarians people who sell their labor for wages

At the heart of Marx's thinking is the idea of **social conflict**, *the struggle between segments of society over valued resources.* Social conflict can, of course, take many forms: Individuals quarrel, colleges have long-standing sports rivalries, and nations sometimes go to war. For Marx, however, the most important type of social conflict was *class conflict* arising from the way a society produces material goods.

Society and Production

Living in the nineteenth century, Marx observed the early decades of industrial capitalism in Europe. This economic system, Marx explained, turned a small part of the population into **capitalists**, *people who own and operate factories and other businesses in pursuit of profits.* A capitalist tries to make a profit by selling a product for more than it costs to produce. Capitalism turns most of the population into industrial workers, whom Marx called **proletarians**, *people who sell their labor for wages.* To Marx, a system of capitalist production always ends up creating conflict between capitalists and workers. To keep profits high, capitalists keep wages low. But workers want higher wages. Since profits and wages come from the same pool of funds, the result is conflict. As Marx saw it, this conflict could end only with the end of capitalism itself.

All societies are composed of **social institutions**, *the major spheres of social life, or societal subsystems, organized to meet human needs.* Examples of social institutions include the economy, the political system, the family, religion, and education. In his analysis of society, Marx argued that one institution—the economy—dominates all the others and defines the character of the entire society. Drawing on the philosophical approach called *materialism,* which says that how humans produce material goods shapes their experiences, Marx believed that the other social institutions all operate in a way that supports a society's economy. Lenski focused on how technology molds a society but, for Marx, it is the economy that forms a society's "real foundation" (1959:43, orig. 1859).

Marx viewed the economic system as society's *infrastructure* (*infra* is Latin, meaning "below"). Other social institutions, including the family, the political system, and religion, are built on this foundation; they form society's *superstructure* and support the economy. Marx's theory is illustrated in Figure 4–1. For example, under capitalism, the legal system protects capitalists' wealth, and the family allows capitalists to pass their property from one generation to the next.

class conflict conflict between entire classes over the distribution of a society's wealth and power

class consciousness workers' recognition of themselves as a class unified in opposition to capitalists and ultimately to capitalism itself

Marx was well aware that most people living in an industrial capitalist system do not recognize how capitalism shapes the operation of their entire society. Most people, in fact, regard the right to own private property or pass it on to their children as "natural." In the same way, many of us tend to see rich people as having "earned" their money through long years of schooling and hard work; we see the poor, on the other hand, as lacking skills and the personal drive to make more of themselves. Marx rejected this type of thinking, calling it **false consciousness**, *explaining social problems as the shortcomings of individuals rather than as the flaws of society.* Marx was saying, in effect, that it is not "people" who make society so unequal but rather the system of capitalist production. False consciousness, he believed, hurts people by hiding the real cause of their problems.

Conflict and History

For Marx, conflict is the engine that drives social change. Sometimes societies change at a slow, *evolutionary* rate. But they may erupt in rapid, *revolutionary* change.

To Marx, early hunters and gatherers formed primitive communist societies. *Communism* is a system in which people commonly own and equally share food and other things they produce. People in hunting and gathering societies do not have much, but they share what they have. In addition, because everyone does the same kind of work, there are no class differences and thus little chance of social conflict.

With technological advance comes social inequality. Among horticultural, pastoral, and early agrarian societies—which Marx lumped together as the "ancient world"—warfare was frequent, and the victors turned their captives into slaves.

Agriculture brings still more wealth to a society's elite but does little for most other people, who labor as serfs and are barely better off than slaves. As Marx saw it, the state supported the feudal system (in which the elite or nobility had all the power), assisted by the church, which claimed that this arrangement reflected the will of God. This is why Marx thought that feudalism was simply "exploitation, veiled by religious and political illusions" (Marx & Engels, 1972:337, orig. 1848).

Gradually, new productive forces started to break down the feudal order. As trade steadily increased, cities grew, and merchants and skilled craftsworkers formed the new capitalist class or *bourgeoisie* (a French word meaning "people of the town"). After 1800, the bourgeoisie also controlled factories, becoming richer and richer so that they soon rivaled the ancient landowning nobility. For their part, the nobles looked down their noses at this upstart "commercial" class, but in time, these capitalists took control of European societies. To Marx's way of thinking, then, new technology was only part of the Industrial Revolution; it also served as a class revolution in which capitalists overthrew the old agrarian elite.

Industrialization also led to the formation of the proletariat. English landowners converted fields once plowed by serfs into grazing land for sheep to produce wool for the textile mills. Forced from the land, millions of people migrated to cities and had little choice but to work in factories. Marx envisioned these workers one day joining together to form a revolutionary class that would overthrow the capitalist system.

Capitalism and Class Conflict

"The history of all hitherto existing society is the history of class struggles." With these words, Marx and his collaborator, Friedrich Engels, began their best-known statement, the *Manifesto of the Communist Party* (1972:335, orig. 1848). Industrial capitalism, like earlier types of society, contains two major social classes: the ruling class, whose members (capitalists or bourgeoisie) own productive property, and the oppressed (proletarians), who sell their labor, reflecting the two basic positions in the productive system. Like masters and slaves in the ancient world and like nobles and serfs in feudal systems, capitalists and proletarians are engaged in class conflict today. Currently, as in the past, one class controls the other as productive property. Marx used the term **class conflict** (and sometimes *class struggle*) to refer to *conflict between entire classes over the distribution of a society's wealth and power.*

Class conflict is nothing new. What distinguishes the conflict in capitalist society, Marx pointed out, is how out in the open it is. Agrarian nobles and serfs, for all their differences, were bound together by traditions and mutual obligations. Industrial capitalism dissolved those ties so that loyalty and honor were replaced by "naked self-interest." Because the proletarians had no personal ties to the capitalists, Marx saw no reason for them to put up with their oppression.

Marx knew that revolution would not come easily. First, workers must *become aware* of their oppression and see capitalism as its true cause. Second, they must *organize and act* to address their problems. This means that false consciousness must be replaced with **class consciousness**, *workers' recognition of themselves as a class unified in opposition to capitalists and ultimately to capitalism itself.* Because the inhumanity of early capitalism was plain for him to see, Marx concluded that industrial workers would soon rise up to destroy this economic system.

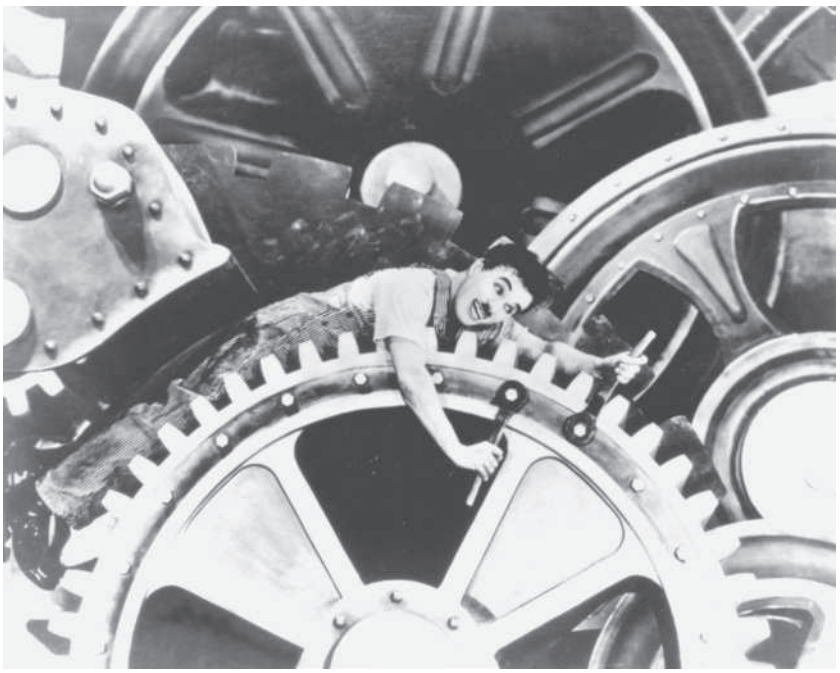
How would the capitalists react? Their wealth made them strong. But Marx saw a weakness in the capitalist armor. Motivated by a desire for personal gain, capitalists feared competition with other capitalists. Marx predicted, therefore, that capitalists would be slow to band together despite their common interests. In addition, he reasoned, capitalists kept employees' wages low in order to maximize profits, which made the workers' misery ever greater. In the long run, Marx believed, capitalists would bring about their own undoing.

Capitalism and Alienation

Marx also condemned capitalist society for producing **alienation**, *the experience of isolation and misery resulting from powerlessness.* To the capitalists, workers are nothing more than a source of labor, to be hired and fired at will. Dehumanized by their jobs (repetitive factory work in the past and processing orders on a computer today), workers find little satisfaction and feel unable to improve their situation. Here we see another contradiction of capitalist society: As people develop technology to gain power over the world, the capitalist economy gains more control over people.

Marx noted four ways in which capitalism alienates workers:

1. **Alienation from the act of working.** Ideally, people work to meet their needs and to develop their personal potential. Capitalism, however, denies workers a say in what they make or how they make it. Further, much of the work is a repetition of routine tasks. The fact that today we replace workers with machines whenever possible would not have surprised Marx. As far as he was concerned, capitalism had turned human beings into machines long ago.
2. **Alienation from the products of work.** The product of work belongs not to workers but to capitalists, who sell it for profit. Thus, Marx reasoned, the more of themselves workers invest in their work, the more they lose.
3. **Alienation from other workers.** Through work, Marx claimed, people build bonds of community. Industrial capitalism, however, makes work competitive rather than cooperative, setting each person apart from everyone else and offering little chance for companionship.
4. **Alienation from human potential.** Industrial capitalism alienates workers from their human potential. Marx argued that a worker "does not fulfill himself in his work but denies himself, has a feeling of misery rather than well-being, does not freely develop his physical and mental energies, but is physically exhausted and mentally debased. The worker, therefore, feels himself to be at home only during his leisure time, whereas at work he feels homeless" (1964: 124–25, orig. 1848). In short, industrial capitalism turns an activity that should express the best qualities in human beings into a dull and dehumanizing experience.



A common fear among thinkers in the early industrial era was that people, now slaves to the new machines, would be stripped of their humanity. No one captured this idea better than the comic actor Charlie Chaplin, who wrote and starred in the 1936 film *Modern Times*.

Marx viewed alienation, in its various forms, as a barrier to social change. But he hoped that industrial workers would overcome their alienation by uniting into a true social class, aware of the cause of their problems and ready to change society.

Revolution

The only way out of the trap of capitalism, Marx argued, is to remake society. He imagined a system of production that could provide for the social needs of all. He called this system *socialism.* Although Marx knew that such a dramatic change would not come easily, he must have been disappointed that he did not live to see workers in England rise up. Still, convinced that capitalism was a social evil, he believed that in time the working majority would realize they held the key to a better future. This change would certainly be revolutionary and perhaps even violent. Marx believed that a socialist society would bring class conflict to an end.

Chapter 10 ("Social Stratification") explains more about changes in industrial-capitalist societies since Marx's time and why the revolution he envisioned never took place. In addition, as Chapter 17 ("Politics and Government") explains, Marx failed to foresee that the revolution he imagined could take the form of repressive regimes, such as Stalin's government in the Soviet Union, that would end up killing tens of millions of people (R. F. Hamilton, 2001). But in his own time, Marx looked toward the future with hope: "The proletarians have nothing to lose but their chains. They have a world to win" (Marx & Engels, 1972:362, orig. 1848).

Max Weber: The Rationalization of Society

4.3 Demonstrate the importance of ideas to the development of human societies.

With a wide-ranging knowledge of law, economics, religion, and history, Max Weber (1864–1920) produced what many experts regard as the greatest individual contribution ever made to sociology. This scholar, born to a prosperous family in Germany, had much to say about how modern society differs from earlier types of social organization.

Weber understood the power of technology, and he shared many of Marx's ideas about social conflict. But he disagreed with Marx's philosophy of materialism. Weber's philosophical approach, called *idealism,* emphasized how human ideas—especially beliefs and values—shape society. He argued that the most important difference among societies is not how people produce things but how people think about the world. In Weber's view, modern society was the product of a new way of thinking.

Weber compared societies in different times and places. To make the comparisons, he relied on the **ideal type**, *an abstract statement of the essential characteristics of any social phenomenon.* Following Weber's approach, for example, we might speak of "preindustrial society" and "industrial society" as ideal types. The use of the word "ideal" does not mean that one or the other is "good" or "best." Nor does an ideal type refer to any actual society. Rather, think of an ideal type as a way of defining a type of society in its pure form. We have already used ideal types in comparing "hunting and gathering societies" with "industrial societies" and "capitalism" with "socialism."

Two Worldviews: Tradition and Rationality

Rather than categorizing societies according to their technology or productive systems, Weber focused on ways that people think about their world. Members of preindustrial societies, Weber explained, are bound by *tradition,* and people in industrial-capitalist societies are guided by *rationality*.

By **tradition**, Weber meant *values and beliefs passed from generation to generation.* In other words, traditional people are guided by the past, and they feel a strong attachment to long-established ways of life. They consider particular actions right and proper mostly because they have been accepted for so long.

People in modern societies, however, favor **rationality**, *a way of thinking that emphasizes deliberate, matter-of-fact calculation of the most efficient way to accomplish a particular task.* Sentimental ties to the past have no place in a rational worldview, and tradition becomes simply one type of information. Typically, modern people think and act on the basis of what they see as the present and future consequences of their choices. They evaluate jobs, schooling, and even relationships in terms of what they put into them and what they expect to receive in return.

Weber viewed both the Industrial Revolution and the development of capitalism as evidence of modern rationality. Such changes are all part of the **rationalization of society**, *the historical change from tradition to rationality as the main type of human thought.* Weber went on to describe modern society as "disenchanted" because scientific thinking has swept away most of people's sentimental ties to the past.

The eagerness to develop new technology and the willingness of people to adopt it as part of their daily lives are

To the outside observer, the trading floor of a stock exchange may look like complete craziness. But Weber saw such activity as a clear expression of modern rationality.